```
> restart; #The best fit for photons Planck's low
> with(PDEtools, dchange):with(plots):
> with(Units):
Automatically loading the Units[Simple] subpackage

> kernelopts(maxdigits);
```

$$38654705646 \tag{1}$$

```
> Digits:=10;
```

$$Digits := 10 \tag{2}$$

```
> params:={k= 1.380649*10^(-23)*Unit(J/K),rb=5.293*10^(-11)*Unit(m),he=
  4.135667697*10^(-15)*Unit(eV*s),
  ec= 1.602176634*10^(-19)*Unit(C),Tq=1.765*10^(-19)*Unit(s),
  c= 299792458*Unit(m/s),T=297*Unit(K),nu1=7.880979442*10^14*Unit(s^(-1)),
  E1=3.259*Unit(eV)};
```

$$params := \left\{E1 = 5.221493608 \times 10^{-19}\ \text{J},\ T = 297\ \text{K},\ Tq = 1.765000000 \times 10^{-19}\ \text{s},\ c = 299792458\ \frac{\text{m}}{\text{s}},\ ec = 1.602176634 \times 10^{-19}\ \text{C},\ he = 6.626070096 \times 10^{-34}\ \frac{\text{m}^2\,\text{kg}}{\text{s}},\ k = 1.380649000 \times 10^{-23}\ \frac{\text{m}^2\,\text{kg}}{\text{s}^2\,\text{K}},\ vl = 7.880979442 \times 10^{14}\ \frac{1}{\text{s}},\ rb = 5.293000000 \times 10^{-11}\ \text{m}\right\} \tag{3}$$

```
> NV:=8*Pi*Tq*nu^2*he*nu/(exp((he*nu-Em)/E0 - 1));
```

$$NV := \frac{8\,\pi\,Tq\,\nu^3\,he}{e^{\frac{he\,\nu - Em}{E0}} - 1} \tag{4}$$

```
> E:=diff(NV,nu);
```

$$E := \frac{24\,\pi\,Tq\,\nu^2\,he}{e^{\frac{he\,\nu - Em}{E0}} - 1} - \frac{8\,\pi\,Tq\,\nu^3\,he^2}{\left(e^{\frac{he\,\nu - Em}{E0}} - 1\right)E0} \tag{5}$$

```
> solve(E=0);
```

$$\{E0 = E0, Em = Em, Tq = Tq, he = he, \nu = 0\},\ \{E0 = E0, Em = Em, Tq = Tq, he = 0, \nu = \nu\},\ \{E0 = E0, Em = Em, Tq = 0, he = he, \nu = \nu\},\ \left\{E0 = \frac{he\,\nu}{3}, Em = Em, Tq = Tq, he = he, \nu = \nu\right\} \tag{6}$$

```
> #Thus, for k=3/2*k*T we have the solution
> c:=299792458;nu1:=7.880979442*10^14;c/nu1;ec:= 1.602176634*10^(-19);E1:=
  3.259*ec;T:=297;Tq:=1.765*10^(-19);h := 6.62607015*10^(-34);he := h/ec;
  E0:=140;Em:=0.0038;
```

$$c := 299792458$$

$$vl := 7.880979442 \times 10^{14}$$

$$3.804000000 \times 10^{-7}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$E1 := 5.221493650 \times 10^{-19}$$

$$T := 297$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$h := 6.626070150 \times 10^{-34}$$

$$he := 4.135667697 \times 10^{-15}$$

$$E0 := 140$$

$$Em := 0.0038 \tag{7}$$

```
> NV:=8*Pi*Tq*nu^2*he*nu/(exp((he*nu-0.038)/(148.5) - 1));
```

$$NV := \frac{1.834552756 \times 10^{-32}\, \nu^3}{e^{2.784961412 \times 10^{-17}\, \nu} - 1.000255892} \tag{8}$$

```
> NV;
```

$$\frac{1.834552756 \times 10^{-32}\, \nu^3}{e^{2.784961412 \times 10^{-17}\, \nu} - 1.000255892} \tag{9}$$

```
> eval(eval(NV,{E(nu)=E__nu1,nu=nu1}),params);
```

$$2.388605408 \times 10^{13} \tag{10}$$

```
> E:=diff(NV,nu);
```

$$E := \frac{5.503658268 \times 10^{-32}\, \nu^2}{e^{2.784961412 \times 10^{-17}\, \nu} - 1.000255892} - \frac{5.109158634 \times 10^{-49}\, \nu^3}{e^{2.784961412 \times 10^{-17}\, \nu} - 1.000255892} \tag{11}$$

```
> solve(E=0);solve(24*Pi*Tq*nu^2*he/exp((he*nu - Em)/E0 - 1) - 8*Pi*Tq*nu^3*
  he^2/(exp((he*nu - Em)/E0 - 1)*E0)=he*nu);
```

$$0.,\ 1.077214207 \times 10^{17}$$

Warning, solutions may have been lost

$$0.,\ 7.439627280 \times 10^{15} - 3.413209702 \times 10^{16}\, I \tag{12}$$

```
> series(24*Pi*Tq*nu^2*he/exp((he*nu - Em)/E0 - 1) - 8*Pi*Tq*nu^3*he^2/(exp(
  (he*nu - Em)/E0 - 1)*E0),nu,9);
```

$$1.496090033 \times 10^{-31}\, \nu^2 - 5.892696401 \times 10^{-48}\, \nu^3 + 1.087956881 \times 10^{-64}\, \nu^4 - 1.285550893 \times 10^{-81}\, \nu^5 + 1.107627350 \times 10^{-98}\, \nu^6 - 7.478822185 \times 10^{-116}\, \nu^7 + 4.142400243 \times 10^{-133}\, \nu^8 + O\left(\nu^9\right) \tag{13}$$

```
> h1:=convert(%,polynom)/(1.6*10^(-18));
```

$$h1 := 9.350562706 \times 10^{-14}\, \nu^2 - 3.682935251 \times 10^{-30}\, \nu^3 + 6.799730506 \times 10^{-47}\, \nu^4 - 8.034693081 \times 10^{-64}\, \nu^5 + 6.922670938 \times 10^{-81}\, \nu^6 - 4.674263866 \times 10^{-98}\, \nu^7 + 2.589000152 \times 10^{-115}\, \nu^8 \tag{14}$$

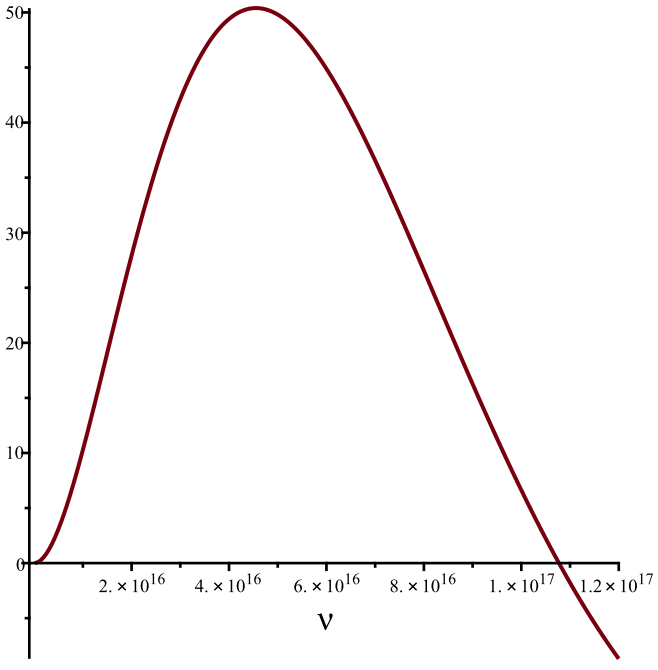
```
> 140*3/he;
```

$$1.015555482 \times 10^{17} \tag{15}$$

```
> lambda_m:=c/%
```

$$lambda_m := 2.952004724 \times 10^{-9} \tag{16}$$

```
> plot(E,nu=10^14..1.2*10^17);#The exact solution
```




```
> evalf(subs(nu=c/(3*10^(-9)),E));
```

$$6.684489485 \tag{17}$$

```
> evalf(c/(3*10^(-9)));1/ec;
```

$$9.993081933 \times 10^{16}$$

$$6.241509074 \times 10^{18} \quad (18)$$

```
> #All we have to do is to fit the constant k=1/2 so for lambda=3.08nm and
  for lambda=380.4nm we have B=3.259 eV
```

```
> evalf(c/(9.993081933*10^16));evalf(c/(380*10^(-9)));evalf(subs(nu=c/
  (380.4*10^(-9)),E));evalf(subs(nu=c/(3.08*10^(-9)),E));
```

$$3.000000001 \times 10^{-9}$$

$$7.889275211 \times 10^{14}$$

$$0.09026023517$$

$$9.088432638 \quad (19)$$

```
> evalf(c/(3.08*10^(-9)));evalf(c/(9.733521364*10^16));
```

$$9.733521364 \times 10^{16}$$

$$3.079999999 \times 10^{-9} \quad (20)$$

```
> evalf(subs(nu=c/(345*10^(-9)),E));evalf(subs(nu=c/(315*10^(-9)),E));evalf
  (subs(nu=c/(10^4*10^(-9)),E));
```

$$0.1094038670$$

$$0.1308314942$$

$$0.0001343429871 \quad (21)$$

```
> evalf(1/2*297);
```

$$148.5000000 \quad (22)$$

```
> restart;#The best fit for photons Planck's low
> E(nu):=Em+E0*ln(1+2.787176283*10^(-17)*nu)+C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.787176283 \times 10^{-17} \nu) + C \nu^2 \quad (23)$$

```
> E1:=subs(nu=9.733521364*10^16,E(nu));
```

$$E1 := Em + E0 \ln(3.712903990) + 9.474143814 \times 10^{33} C \quad (24)$$

```
> E2:=subs(nu=7.880979442*10^14,E(nu));
```

$$E2 := Em + E0 \ln(1.021965679) + 6.210983697 \times 10^{29} C \quad (25)$$

```
> E3:=subs(nu=8.689636465*10^14,E(nu));
```

$$E3 := Em + E0 \ln(1.024219549) + 7.550978189 \times 10^{29} C \quad (26)$$

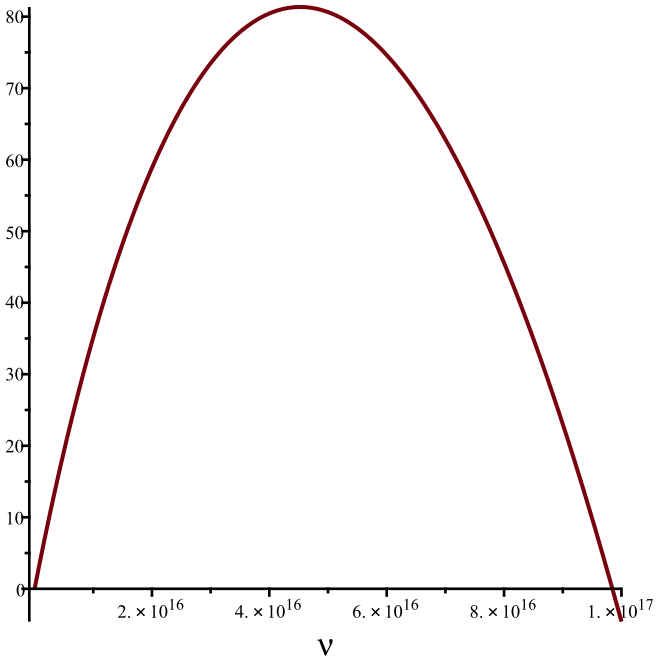
```
> solve({E1=3.259,E2=3.259,E3=3.59});
```

$$\{C = -2.063162888 \times 10^{-32}, E0 = 151.5047392, Em = -0.02006692051\} \quad (27)$$

```
> E(nu):=subs(Em=-0.02006692051,E0=151.5047392,C=-2.063162888*10^(-32),E(nu)
  );
```

$$E(\nu) := -0.02006692051 + 151.5047392 \ln(1 + 2.787176283 \times 10^{-17} \nu) - 2.063162888 \times 10^{-32} \nu^2 \quad (28)$$

```
> plot(E(nu),nu=10^13..10^17);
```




```
> h := 6.6260701500*10^(-34);ec := 1.6021766340*10^(-19);rb:=5.239*10^(-13);
  c := 299792458;T := 273 + 24;k := 1.3806490000*10^(-23);Tq:= 1.765*10^
  (-19);nu0:=evalf(c/(3.804*10^(-7)));nu1:=evalf(c/(3.08*10^(-9)));nu2:=
  evalf(c/(3.135*10^(-7)))
```

$$h := 6.626070150 \times 10^{-34}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$rb := 5.239000000 \times 10^{-13}$$

$$c := 299792458$$

$$T := 297$$

$$k := 1.380649000 \times 10^{-23}$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$v0 := 7.880979442 \times 10^{14}$$

$$v1 := 9.733521364 \times 10^{16}$$

$$v2 := 9.562757830 \times 10^{14} \quad (29)$$

```
> evalf(subs(nu=7.880979442*10^14,E(nu)));
```

$$3.258999999 \quad (30)$$

```
> evalf(subs(nu=8.689636465*10^14,E(nu)));
```

$$3.590000000 \quad (31)$$

```
> evalf(subs(nu=9.733521364*10^16,E(nu)));
```

$$3.2590000 \quad (32)$$

```
> evalf(subs(nu=nu2,E(nu)));
```

$$3.946259923 \quad (33)$$

```
> solve(-0.02006692051 + 151.5047392*ln(1 + 2.787176283*10^(-17)*nu) -
  2.063162888*10^(-32)*nu^2=5.28,nu)
Warning, solutions may have been lost
> evalf(subs(nu=5*10^16,E(nu)));
```

$$80.63322078 \quad (34)$$

```
> evalf(subs(nu=0.984563*10^17,E(nu)));c/(0.984563*10^17);
```

$$0.0000560$$

$$3.044929150 \times 10^{-9} \quad (35)$$

```
> restart;
> with(PDEtools, dchange):with(plots):
> with(Units):
```

```
Automatically loading the Units[Simple] subpackage
```

```
> kernelopts(maxdigits);
```

$$38654705646 \tag{36}$$

```
> Digits:=10;
```

$$Digits := 10 \tag{37}$$

```
> params:={k= 1.380649*10^(-23)*Unit(J/K),rb=5.293*10^(-11)*Unit(m),he=
  4.135667697*10^(-15)*Unit(eV*s),
  ec= 1.602176634*10^(-19)*Unit(C),Tq=1.765*10^(-19)*Unit(s),
  c= 299792458*Unit(m/s),T=297*Unit(K),nu1=7.880979442*10^14*Unit(s^(-1)),
  E1=3.259*Unit(eV)};
```

$$params := \left\{E1 = 5.221493608 \times 10^{-19}\ \text{J},\ T = 297\ \text{K},\ Tq = 1.765000000 \times 10^{-19}\ \text{s},\ c = 299792458\ \frac{\text{m}}{\text{s}},\ ec = 1.602176634 \times 10^{-19}\ \text{C},\ he = 6.626070096 \times 10^{-34}\ \frac{\text{m}^2\,\text{kg}}{\text{s}},\ k = 1.380649000 \times 10^{-23}\ \frac{\text{m}^2\,\text{kg}}{\text{s}^2\,\text{K}},\ vl = 7.880979442 \times 10^{14}\ \frac{1}{\text{s}},\ rb = 5.293000000 \times 10^{-11}\ \text{m}\right\} \tag{38}$$

```
> NV:=8*Pi*Tq*nu^2*h(nu)*nu/(exp(h(nu)*nu/(k*T) - 1));
```

$$NV := \frac{8\,\pi\,Tq\,\nu^3\,h(\nu)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} \tag{39}$$

```
> E:=nu*h(nu);
```

$$E := \nu\,h(\nu) \tag{40}$$

```
> ic:=h(7.889275211*10^14)=4.135667697*10^(-15);
```

$$ic := h(7.889275211 \times 10^{14}) = 4.135667697 \times 10^{-15} \tag{41}$$

```
> de:=diff(NV,nu)-h(nu)*nu;
```

$$de := \frac{24\,\pi\,Tq\,\nu^2\,h(\nu)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} + \frac{8\,\pi\,Tq\,\nu^3\left(\frac{d}{d\nu}\,h(\nu)\right)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} - \frac{8\,\pi\,Tq\,\nu^3\,h(\nu)\left(\frac{\left(\frac{d}{d\nu}\,h(\nu)\right)\nu}{kT} + \frac{h(\nu)}{kT}\right)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} - \nu\,h(\nu) \tag{42}$$

```
> dsolve({de});
```

$$[\varnothing] \tag{43}$$

```
> #Thus, for k=3/2*k*T we have the solution
> c:=299792458;nu1:=7.880979442*10^14;c/nu1;ec:= 1.602176634*10^(-19);k:=
  1/2;E1:=3.259*ec;T:=297;Tq:=1.765*10^(-19);he := h/ec;
```

$$c := 299792458$$

$$vl := 7.880979442 \times 10^{14}$$

$$3.804000000 \times 10^{-7}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$k := \frac{1}{2}$$

$$E1 := 5.221493650 \times 10^{-19}$$

$$T := 297$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$he := 6.241509074 \times 10^{18}\,h \tag{44}$$

```
> NV:=8*Pi*Tq*nu^2*he(nu)*nu/(exp(he(nu)*nu/(k*T) - 1));
```

$$NV := \frac{27.68689003\,\nu^3\,h(\nu)}{e^{4.203036413\times 10^{16}\,\nu\,h(\nu)} - 1} \tag{45}$$

```
> ic1:=h(9.489275211*10^16)=4.135667697*10^(-17);
```

$$ic1 := h(9.489275211\times 10^{16}) = 4.135667697\times 10^{-17} \tag{46}$$

```
> de:=diff(NV,nu)-h(nu)*nu;
```

$$de := \frac{83.06067009\,\nu^2\,h(\nu)}{e^{4.203036413\times 10^{16}\,\nu\,h(\nu)} - 1} + \frac{27.68689003\,\nu^3\left(\frac{d}{d\nu}\,h(\nu)\right)}{e^{4.203036413\times 10^{16}\,\nu\,h(\nu)} - 1} - \frac{27.68689003\,\nu^3\,h(\nu)\left(4.203036413\times 10^{16}\,h(\nu) + 4.203036413\times 10^{16}\left(\frac{d}{d\nu}\,h(\nu)\right)\nu\right)}{e^{4.203036413\times 10^{16}\,\nu\,h(\nu)} - 1} - \nu\,h(\nu) \tag{47}$$

```
> evalf(simplify(dsolve({de,ic1},h(nu),series)));
```

$$h(\nu) = 4.135667697\times 10^{-17} - 4.060327648\times 10^{71635144604154865}\left(\nu - 9.489275211\times 10^{16}\right) + 3.287671071 \times 10^{143270289208309764}\left(\nu - 9.489275211\times 10^{16}\right)^2 - 3.549395354\times 10^{214905433812464663}\left(\nu - 9.489275211\times 10^{16}\right)^3 + 4.310949299\times 10^{286540578416619562}\left(\nu - 9.489275211\times 10^{16}\right)^4 - 5.584961425\times 10^{358175723020774461}\left(\nu - 9.489275211\times 10^{16}\right)^5 + O\left(\left(\nu - 9.489275211\times 10^{16}\right)^6\right) \tag{48}$$

```
> h0(nu):=convert(%,polynom);
```

$$h0(\nu) := h(\nu) = 3.852956650\times 10^{71635144604154882} - 4.060327648\times 10^{71635144604154865}\,\nu + 3.287671071 \times 10^{143270289208309764}\left(\nu - 9.489275211\times 10^{16}\right)^2 - 3.549395354\times 10^{214905433812464663}\left(\nu - 9.489275211\times 10^{16}\right)^3 + 4.310949299\times 10^{286540578416619562}\left(\nu - 9.489275211\times 10^{16}\right)^4 - 5.584961425\times 10^{358175723020774461}\left(\nu - 9.489275211\times 10^{16}\right)^5 \tag{49}$$

```
> restart;
> dsolve(E(nu)=diff(nu^2*E(nu)/(exp(E(nu)/2)-1),nu));
> E1(nu):=-16*E0*exp(2/(A*nu))*(-A^2*nu^2*exp(2/(A*nu))/8 - A*nu*exp(2/(A*
  nu))/4 - Ei(1, abs(-2/(A*nu)))/2)/(nu^2*A^2) + 4*E0*C;
```

$$E1(\nu) := -\frac{16\,E0\,e^{\frac{2}{A\nu}}\left(-\frac{A^2\nu^2 e^{\frac{2}{A\nu}}}{8} - \frac{A\nu e^{\frac{2}{A\nu}}}{4} - \frac{\mathrm{Ei}_1\left(\frac{2}{|A\nu|}\right)}{2}\right)}{\nu^2 A^2} + 4\,E0\,C \tag{50}$$

```
> E(nu):=4*E0/nu^2*exp(2/nu)*int(nu*exp(2/nu),nu)+4*E0*C;
```

$$E(\nu) := \frac{4\,E0\,e^{\frac{2}{\nu}}\left(\frac{\nu^2 e^{\frac{2}{\nu}}}{2} + \nu e^{\frac{2}{\nu}} + 2\,\mathrm{Ei}_1\left(-\frac{2}{\nu}\right)\right)}{\nu^2} + 4\,E0\,C \tag{51}$$

```
> E(nu) := 4*E0*exp(2/nu)*(nu^2*exp(2/nu)/2 + nu*exp(2/nu) + 2*Ei(1,abs
  (-2/nu)))/nu^2 + 4*E0*C
```

$$E(\nu) := \frac{4\,E0\,e^{\frac{2}{\nu}}\left(\frac{\nu^2 e^{\frac{2}{\nu}}}{2} + \nu e^{\frac{2}{\nu}} + 2\,\mathrm{Ei}_1\left(\frac{2}{|\nu|}\right)\right)}{\nu^2} + 4\,E0\,C \tag{52}$$

```
> E0:=14;E(nu);C:=1;A:=1.765^2*10^(-19);
```

$$E0 := 14$$

$$\frac{4\,E0\,e^{\frac{2}{\nu}}\left(\frac{\nu^2 e^{\frac{2}{\nu}}}{2}+\nu\,e^{\frac{2}{\nu}}+2\,\mathrm{Ei}_1\left(\frac{2}{|\nu|}\right)\right)}{\nu^2}+4\,E0\,C$$

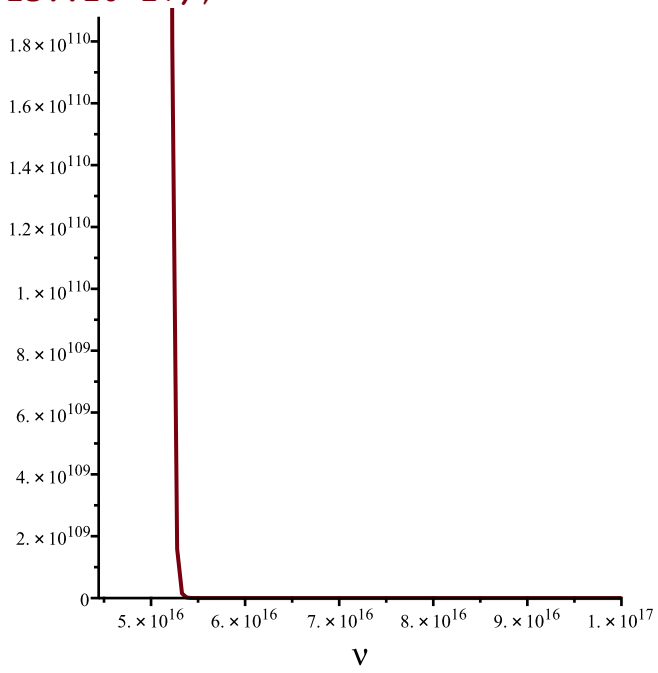
$$C := 1$$

$$A := 3.115225000\times 10^{-19} \tag{53}$$

```
> E1(nu)
```

$$-\frac{16\,E0\,e^{\frac{2}{A\nu}}\left(-\frac{A^2\nu^2 e^{\frac{2}{A\nu}}}{8}-\frac{A\nu\,e^{\frac{2}{A\nu}}}{4}-\frac{\mathrm{Ei}_1\left(\frac{2}{|A\nu|}\right)}{2}\right)}{\nu^2 A^2}+4\,E0\,C \tag{54}$$

```
> plot(E1(nu),nu = 10^13..10^17);
```




```
>
> abs(diff(E(nu),nu));
```

$$4\left|-\frac{2\,E0\,e^{\frac{2}{\nu}}\left(\frac{\nu^2 e^{\frac{2}{\nu}}}{2}+\nu\,e^{\frac{2}{\nu}}+2\,\mathrm{Ei}_1\left(\frac{2}{|\nu|}\right)\right)}{\nu^4}+\frac{E0\,e^{\frac{2}{\nu}}\left(\nu\,e^{\frac{2}{\nu}}-\frac{2\,e^{\frac{2}{\nu}}}{\nu}+\frac{2\,\mathrm{abs}(1,\nu)\,e^{-\frac{2}{|\nu|}}}{|\nu|}\right)}{\nu^2}\right.$$
$$\left.-\frac{2\,E0\,e^{\frac{2}{\nu}}\left(\frac{\nu^2 e^{\frac{2}{\nu}}}{2}+\nu\,e^{\frac{2}{\nu}}+2\,\mathrm{Ei}_1\left(\frac{2}{|\nu|}\right)\right)}{\nu^3}\right| \tag{55}$$

```
> evalf(c/(3.08*10^(-9)));
```

$$3.246753247\times 10^{8}\,c \tag{56}$$

```
> #All we have to do is to fit the constant k=1/2 so for lambda=3.08nm and
  for lambda=380.4nm we have E=3.259 eV
> h:=3.259/(9.733521364*10^16);
```

$$h := 3.348222990\times 10^{-17} \tag{57}$$

```
> k=1.380649*297/1.6*10^(-5);
```

$$k = 0.002562829706 \tag{58}$$

```
> k=1.380649*10^(-23)*297/(1.60217*10^(-19));
```

$$k = 0.02559358576 \tag{59}$$

```
> restart;
> E(nu):=E0*ln(1+16*Pi^2*Tq*nu)+C*nu^2;
```

$$E(\nu) := E0 \ln\left(16 \pi^2 Tq \nu + 1\right) + C \nu^2 \tag{60}$$

```
> Tq:=1.765*10^(-19);16*Tq*Pi^2;ERy := 13.605693122990;
  h := 6.626070150*10^(-34);
  ec := 1.602176634*10^(-19);
  h/ec;
  c := 299792458;c/(3.08*10^(-9));c/(3.804*10^(-7));
```

$$Tq := 1.765000000 \times 10^{-19}$$

$$2.787176283 \times 10^{-17}$$

$$ERy := 13.605693122990$$

$$h := 6.626070150 \times 10^{-34}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$4.135667697 \times 10^{-15}$$

$$c := 299792458$$

$$9.733521364 \times 10^{16}$$

$$7.880979442 \times 10^{14} \tag{61}$$

```
> E(nu):=E0*ln(1+2.787176283*10^(-17)*nu)+ C*nu^2;
```

$$E(\nu) := E0 \ln\left(1 + 2.787176283 \times 10^{-17} \nu\right) + C \nu^2 \tag{62}$$

```
> solve(3.259=subs(nu=9.733521364*10^16,E(nu)),E0);
```

$$-7.222168329 \times 10^{33} C + 2.484345503 \tag{63}$$

```
> solve(3.259=subs(nu=7.880979442*10^14,E(nu)),E0);
```

$$-2.858528029 \times 10^{31} C + 149.9914233 \tag{64}$$

```
> solve(-7.222168329*10^33*C + 2.484345503=-2.858528029*10^31*C +
  149.9914233,C);
```

$$-2.050536941 \times 10^{-32} \tag{65}$$

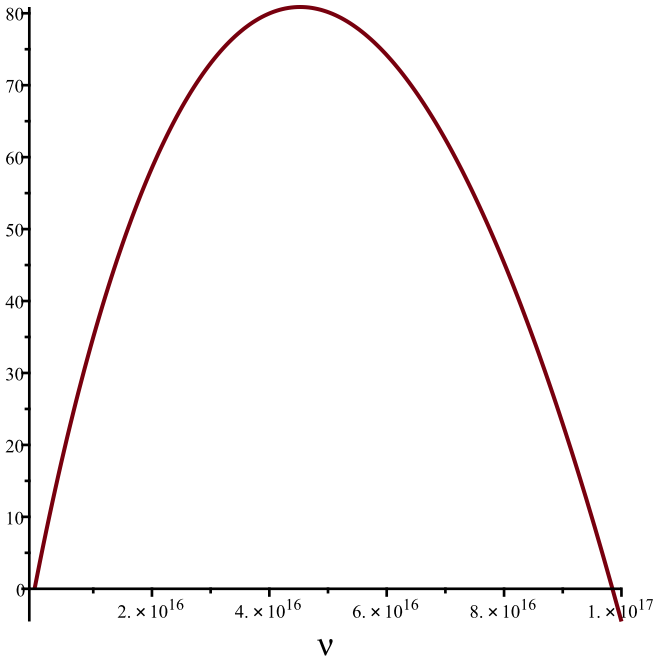
```
> E0:=-2.858528029*10^31*(-2.050536941*10^(-32)) + 149.9914233;
```

$$E0 := 150.5775750 \tag{66}$$

```
> E(nu):=150.5775750*ln(1+2.787176283*10^(-17)*nu)-2.050536941*10^(-32)*
  nu^2;
```

$$E(\nu) := 150.5775750 \ln\left(1 + 2.787176283 \times 10^{-17} \nu\right) - 2.050536941 \times 10^{-32} \nu^2 \tag{67}$$

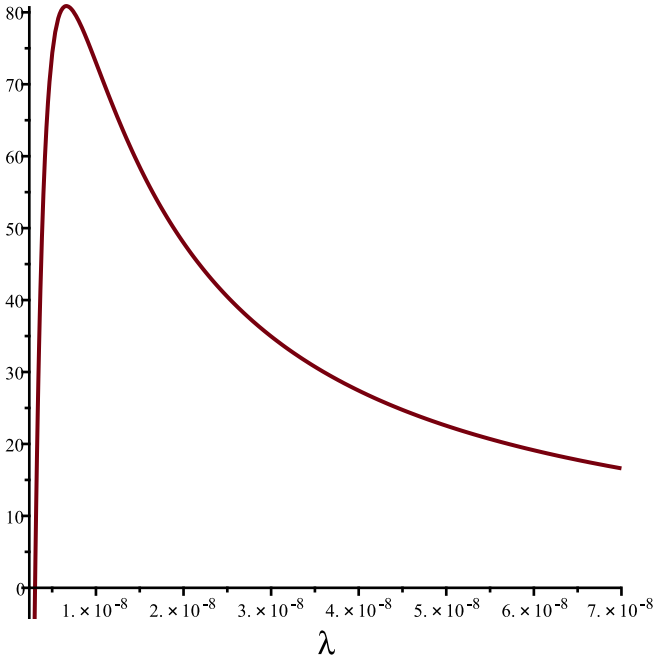
```
> plot(150.5775750*ln(1 + 2.787176283*10^(-17)*nu) - 2.050536941*10^(-32)*
  nu^2,nu=10^13..10^17);
```




```
> E1(lambda):=subs(nu=c/lambda,E(nu));
```

$$EI(\lambda) := 150.5775750 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) - \frac{1.842930695 \times 10^{-15}}{\lambda^2} \tag{68}$$

```
> plot(E1(lambda),lambda=7*10^(-8)..3*10^(-9));
```




```
> E1(lambda);
```

$$150.5775750 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) - \frac{1.842930695 \times 10^{-15}}{\lambda^2} \tag{69}$$

```
>
```